

KeyNav: Robust multi-modal quadruped state estimation for automated industrial inspection

Miguel Castellón, Hilario Tomé

Abstract—We present a robust multi-modal sensor fusion framework for accurate odometry estimation tailored to quadruped robots operating in challenging industrial environments such as chemical plants. Our method integrates data from lidar, stereo cameras, multiple IMUs, and kinematic sensors within a factor graph optimization framework. By leveraging complementary sensor modalities, the system maintains reliable performance even when individual sensors provide degraded data. Extensive evaluation in simulation and real-world deployments demonstrates that our approach achieves high precision, enables navigation in confined and dynamic spaces, and produces detailed 3D maps, bringing increased reliability and safety to autonomous industrial inspections.

Index Terms—SLAM, odometry, lidar, multi-modal.

I. INTRODUCTION

Robust and precise odometry is fundamental for safe and effective deployment of quadruped robots in industrial environments, where sensor degradation and environmental uncertainties are common. Traditional visual-inertial odometry (VIO) and LiDAR-based methods (LIO) often underperform or even fail in such real-world scenarios due to a lack of reliable features, occlusions and structural ambiguities, which lead to ill-posed optimization problems. Our approach addresses these challenges by introducing a probabilistic, tightly coupled multi-modal fusion pipeline that ensures consistent state estimation even under sensor degeneration. Our method has been extensively validated with simulated and real data, and has powered the deployment of Keyper in our customer plants (see fig. 1).

A. Platform description

Keyper is a quadruped robot designed for automated industrial inspections. The exteroceptive sensors with which it is equipped are a lidar and five stereo camera pairs. It also includes proprioceptive sensors, such as IMUs and information on joint states and foot contact status.

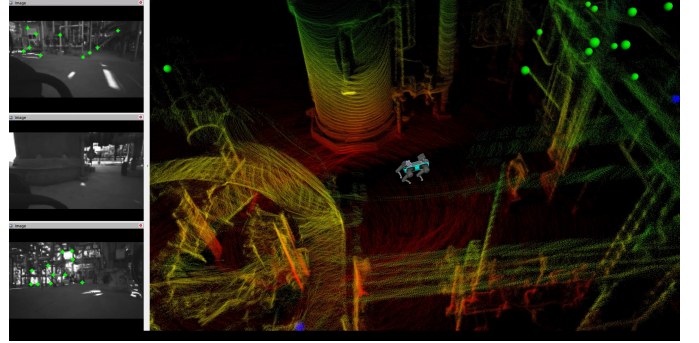
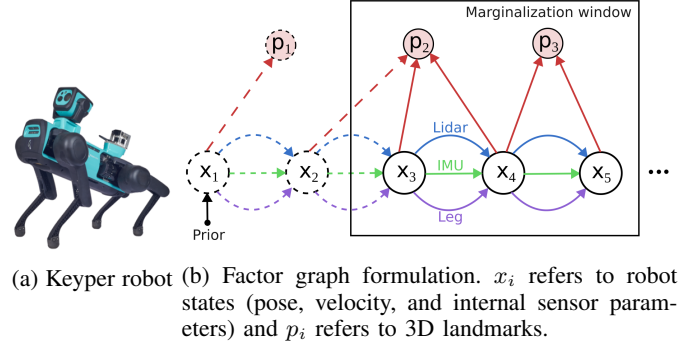
In addition, Keyper incorporates advanced imaging sensors—such as a zoom camera for detailed inspections, a thermal camera for temperature mapping, and a high-fidelity microphone for acoustic monitoring.

B. Deployment

The deployment of Keyper is based on a teach-and-repeat approach. The user can easily record a mission by teleoperating the robot to positions of interest, from where tasks such as

M. Castellón and H. Tomé are with Keybotic SL (<http://keybotic.com>), Barcelona, Spain.

E-mails: miguel.castillon@keybotic.com and hilario.tome@keybotic.com



(c) Odometry estimation. Full video: <https://youtu.be/Pi1MpdA9X4Q>

Fig. 1: Our multi-modal state estimation pipeline is the backbone of our autonomous navigation stack, which allows Keyper to conduct automated inspection tasks in real industrial plants.

gauge inspection or thermal monitoring need to be performed. During this phase, KeyNav creates a 3D representation of the environment by combining odometry outputs with loop closure constraints. After that, Keyper can automatically repeat the programmed inspection route by autonomously navigating to the desired positions thanks to its global map localization method.

C. Requirements

In order to safely and reliably navigate real industrial environments, our state estimation pipeline must meet strict performance requirements. In particular, the odometry module needs to be light enough to produce online estimations at 20 Hz while leaving enough CPU resources free for other modules, such as gait control and mission-specific processing. Moreover, the quality of its output estimation must respect industrial standards to avoid potentially dangerous situations for personnel and equipment.

D. Perception challenges in real environments

Industrial plants are complex semi-structured environments. A mobile robot deployed in them often receives degenerate sensor data which may corrupt the estimation if not properly handled.

One typical failure mode of lidar odometry pipelines are vast flat or corridor-like areas where the normals of lidar data do not constrain all directions, such as in fig. 2. This usually leads to ill-posed optimization problems.

Regarding camera input, disadvantageous lighting conditions happen relatively often in outdoor environments. As shown in fig. 3, when direct sunlight impacts on the camera, the contrast in the image is drastically reduced, leading to large over- or underexposed areas from which reliable features are difficult to track.

Moreover, both lidar and visual data can be corrupted by obstacles moving around the robot (see fig. 4). If these outliers are not properly detected and handled, the movement of such obstacles can negatively influence the output of the state estimation.

Finally, kinematic information relies on detecting non-slippery floor contacts. In real operation, where the robot must traverse uneven or slippery terrain, the reliability of contact detectors drops, causing kinematic-only odometry estimation to quickly accumulate large drifts.

II. METHOD

A. Odometry

Our multi-modal state estimation pipeline is divided in parallel threads which process sensor data (referred to as frontends). All sensor frontends are synchronized by timer-based queries and add new data to the backend, which optimizes the problem and publishes its result.

The problem is formulated as a factor graph and solved with Bayes-Tree-based iSAM2 [1] (GTSAM [2]) with a marginalization window (see fig. 1b). The optimized variables include robot poses, velocities, internal parameters of sensors, and 3D landmarks. These variables are related to each other by different types of factors.

1) *IMU factors*: High-rate linear acceleration and angular velocity measurements coming from each IMU are preintegrated as in [3], which creates one factor that constrains the pose, velocity and IMU biases of two consecutive states.

2) *Lidar factors*: Our lidar frontend is based on a degeneracy-aware version of [4] that is able to inform the backend optimization of underconstrained degrees of freedom. We incrementally update a 3D local map with undistorted scan data and create a factor with the current point-to-plane distances. Then, we constrain the optimization in the non-observable directions by setting the corresponding Hessian elements to zero. In comparison with ICP-based odometry methods such as [5], including dense lidar data rather than a single displacement factor between states produces more reliable results without consuming too many resources.

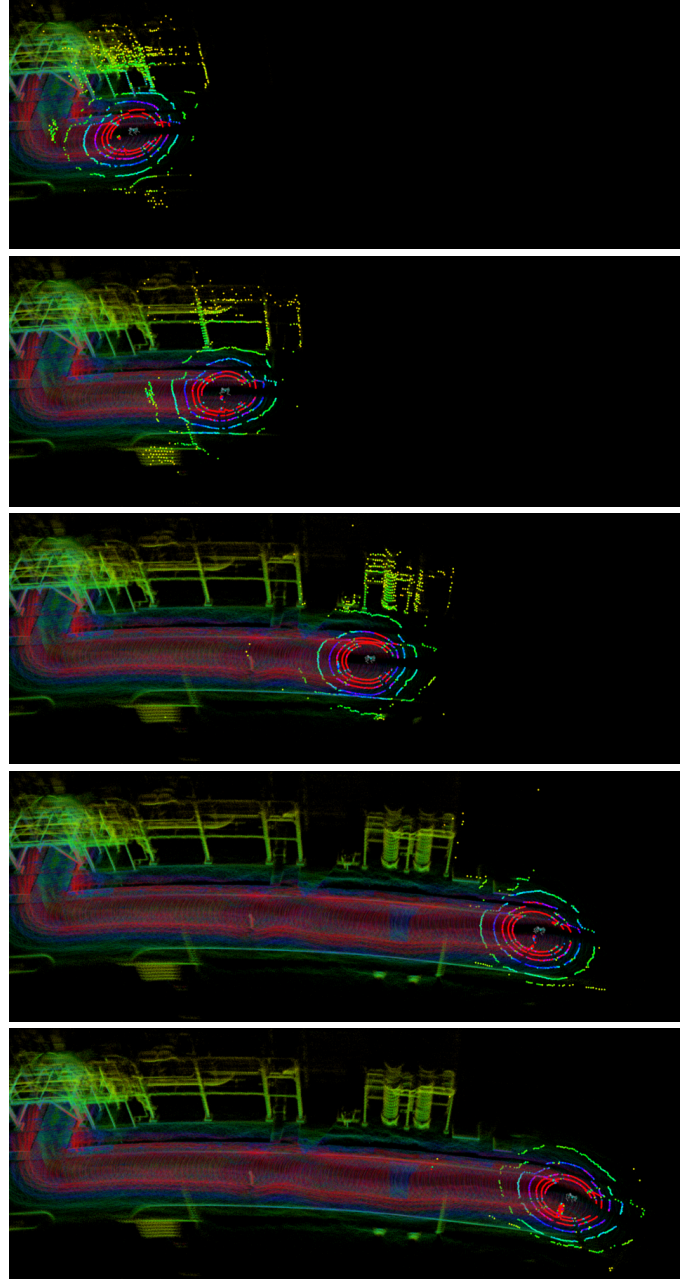


Fig. 2: Sequence of 3D map creation while operating in a real chemical plant. The output odometry estimation stays reliably correct without hard sensor switching by using adequate uncertainty characterization. This is a typical failure case of LIO pipelines because the coplanarity of almost all lidar points leaves 3 degrees of freedom underconstrained.

3) *Visual factors*: Stereo camera factors try to minimize the difference between the expected and the real feature measurements on image space. Following [6], far landmarks are parameterized with inverse depth coordinates to better represent the uncertainty around their 3D positions, which are also optimized. Identifying outliers caused by moving obstacles is crucial; therefore, we discard landmark observations that deviate significantly from their expected position. Moreover, we use a robust cost function to reduce the influence of subtle

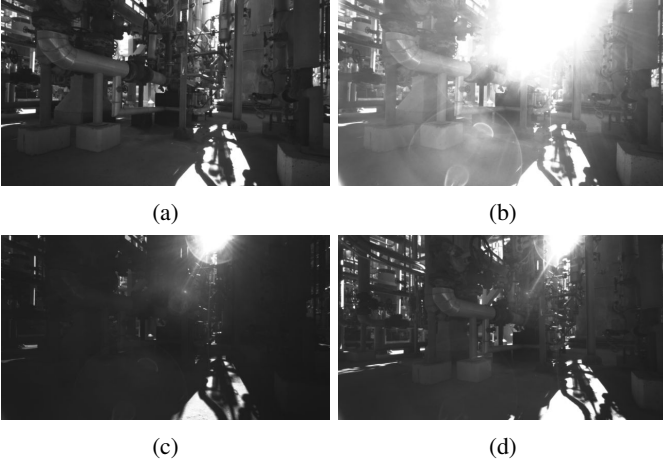


Fig. 3: Sequence of over- and underexposed images. Industrial plants typically have large outdoor areas with non-ideal lighting conditions. Similarly to feature-less areas, this typically makes VIO pipelines fail because of a lack of trackable features.

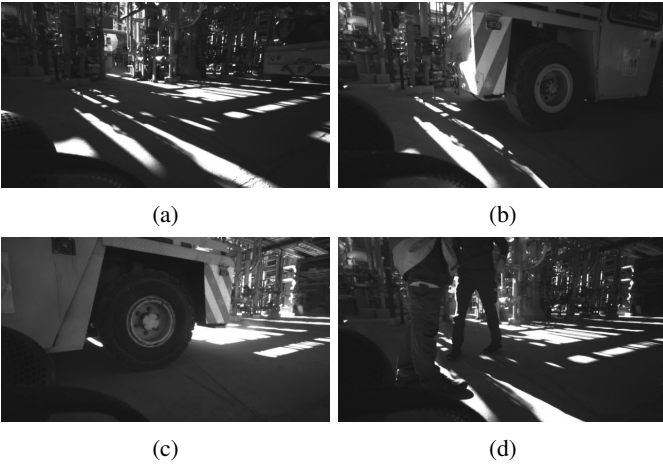
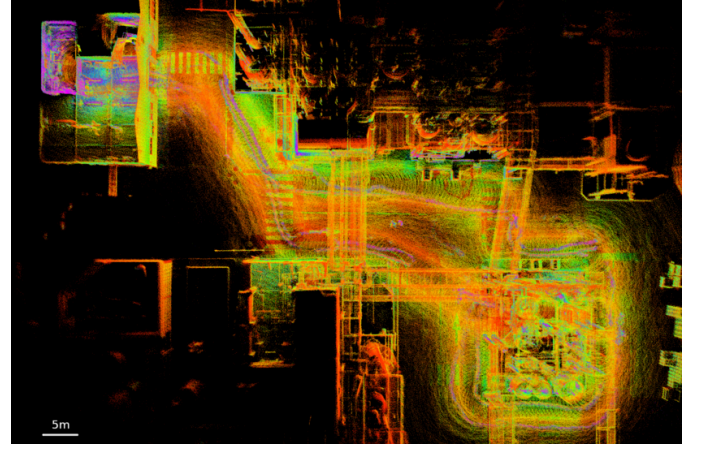


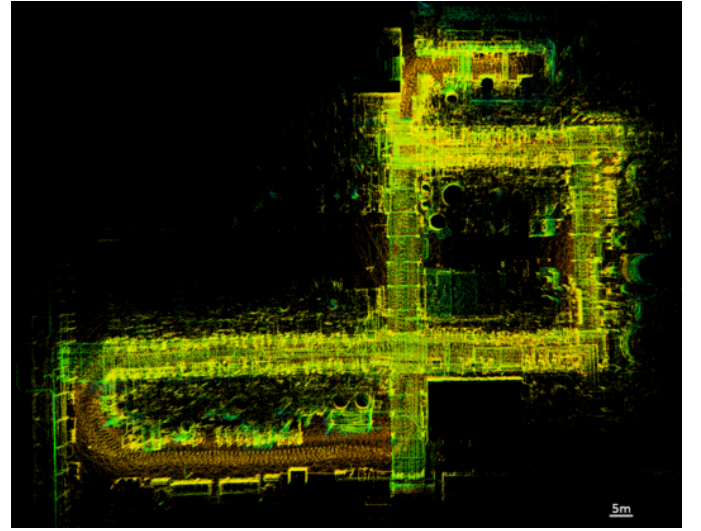
Fig. 4: Sequence of camera frames with moving obstacles (a vehicle and people) covering large parts of its field of view while autonomously operating in a real chemical plant. This is typically a challenge for VIO pipelines because most of the detected features have non-zero velocity, which violates the static-world assumption. Our odometry pipeline identifies outliers robustly.

outliers that, while not extreme enough to trigger rejection, can still degrade the optimization performance.

4) *Leg factors*: Kinematic information is merged with contact status in a similar way to [7], in which velocity measurements from legs with non-slipping contacts are pre-integrated to form displacement constraints. We reformulated that method to add a multiplicative bias, which better fits our observations. In the near future, we will consider the more formally correct approach proposed in [8]. This method avoids factor correlation that emerges from using the same gyroscope data for both IMU and kinematic factors.



(a)



(b)

Fig. 5: Examples of 3D mapped industrial plants where Keyper is deployed for automated inspection tasks.

B. Multi-modal mapping and localization

There are other modules that consume odometry data to achieve autonomous navigation. In particular, our mapping pipeline uses those estimates along with loop closure information to run a low-frequency bundle adjustment method on the robot. The loop closure module uses the probabilistic odometry output to find global matches of visual and lidar data. The optimized lidar and visual data is finally serialized and ready to be used by the multi-modal localization module to perform autonomous navigation.

III. RESULTS

KeyNav successfully powers the automated inspection capabilities of Keyper in customer plants. Examples of resulting 3D maps of such plants are shown in fig. 5. These maps are created and optimized online on the robot and do not require any manual steps, so that Keyper can start executing the recorded missions right after the teaching phase.

IV. CONCLUSIONS

In conclusion, our multi-modal sensor fusion framework significantly enhances the reliability and robustness of quadruped state estimation for industrial inspection tasks. By tightly integrating LiDAR, stereo vision, inertial, and kinematic data within a factor graph optimization framework, the system overcomes individual sensor limitations and meets demanding industrial safety standards. Autonomous navigation is critical for enabling high-value tasks such as thermal monitoring, gauge reading, and gas leak detection, and it must be robust against changing conditions, such as day and night operation. Future work will explore further optimization of computational resources, extension to even more dynamic environments, and integration of additional sensor modalities to broaden the system's applicability.

ACKNOWLEDGMENTS

We would like to thank all members in the Keybotic team across all departments for their extraordinary contributions to the development of Keyper.

REFERENCES

- [1] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. Leonard, and F. Dellaert, "isam2: Incremental smoothing and mapping with fluid relinearization and incremental variable reordering," in *2011 IEEE International Conference on Robotics and Automation*, 2011, pp. 3281–3288.
- [2] F. Dellaert, M. Kaess *et al.*, "Factor graphs for robot perception," *Foundations and Trends® in Robotics*, vol. 6, no. 1-2, pp. 1–139, 2017.
- [3] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, "On-manifold preintegration for real-time visual-inertial odometry," *IEEE Transactions on Robotics*, vol. 33, no. 1, pp. 1–21, 2016.
- [4] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "Fast-lid2: Fast direct lidar-inertial odometry," 2021. [Online]. Available: <https://arxiv.org/abs/2107.06829>
- [5] T. Tuna, J. Nubert, Y. Nava, S. Khattak, and M. Hutter, "X-icp: Localizability-aware lidar registration for robust localization in extreme environments," 2024. [Online]. Available: <https://arxiv.org/abs/2211.16335>
- [6] J. Civera, A. J. Davison, and J. M. Montiel, "Inverse depth parametrization for monocular slam," *IEEE transactions on robotics*, vol. 24, no. 5, pp. 932–945, 2008.
- [7] D. Wisth, M. Camurri, and M. Fallon, "Vilens: Visual, inertial, lidar, and leg odometry for all-terrain legged robots," *IEEE Transactions on Robotics*, vol. 39, no. 1, pp. 309–326, 2023.
- [8] P. Vial, J. Solà, N. Palomeras, and M. Carreras, "On lie group imu and linear velocity preintegration for autonomous navigation considering the earth rotation compensation," *IEEE Transactions on Robotics*, vol. 41, pp. 1346–1364, 2025.